10.TRANSFORM TECHNIQUES AND SYSTEM REPRESENTATION

Objective

This experiment is divided into four simulations:

1. Pole Zero Plot
2. Z-Transform and Inverse Z-Transform

Experiment 10.1

Pole Zero Plot

Aim:

The Aim of this experiment is to understand and analyze the pole-zero plot of a discrete-time system using Scilab. The pole-zero plot is a fundamental tool in digital signal processing (DSP) and control systems, used to visualize the locations of poles and zeros of a system's transfer function in the complex plane.

Program:

```
% Test Case: Manual Pole-Zero Plot
clc;
clear;
close all;
% Define the coefficients of the transfer function
b = [1 -0.5];
a = [1 -1.2 0.32];
% Numerator coefficients (zeros)
```

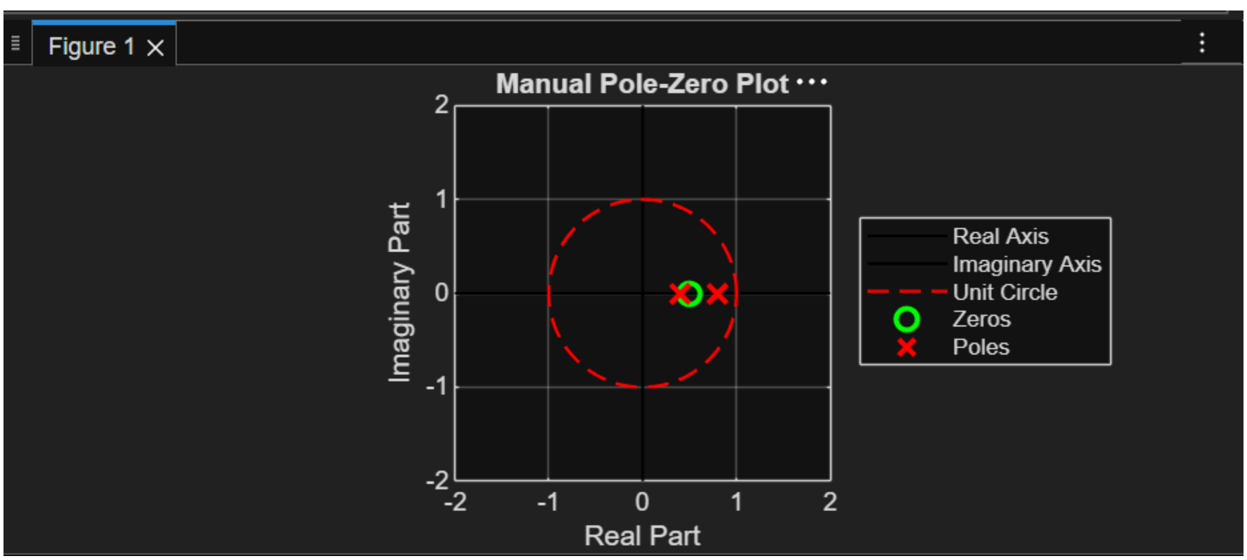
```
% Denominator coefficients (poles)
% Calculate zeros and poles
z = roots(b);
p = roots(a);
% Create a new figure
figure;
% Plot real axis
plot([-2 2], [0 0], 'k-', 'LineWidth', 1);
hold on;
% Plot imaginary axis
plot([0 0], [-2 2], 'k-', 'LineWidth', 1);
% Plot unit circle
theta = linspace(0, 2*pi, 200);
unit_circle_x = cos(theta);
unit_circle_y = sin(theta);
plot(unit_circle_x, unit_circle_y, 'r--', 'LineWidth', 1.2);
% Plot zeros
plot(real(z), imag(z), 'go', ...
'MarkerSize', 8, 'LineWidth', 2);
% Plot poles
plot(real(p), imag(p), 'rx', ...
'MarkerSize', 10, 'LineWidth', 2);
% Customize plot
xlabel('Real Part');
```

```
ylabel('Imaginary Part');
title('Manual Pole-Zero Plot');
legend('Real Axis', 'Imaginary Axis', ...
'Unit Circle', 'Zeros', 'Poles', ...
'Location', 'northwest');
axis equal;
xlim([-2 2]);
ylim([-2 2]);
grid on;
hold off;
```

Expected Results:

- Poles: The poles will be plotted as red crosses ('x') on the complex plane.
- Zeros: The zeros will be plotted as green circles ('o') on the complex plane.
- Unit Circle: The unit circle is plotted in the complex plane to help assess the stability of the system.

Output Waveform:



Results:

- Stability: Check the location of the poles relative to the unit circle. If all poles are inside the unit circle, the system is stable; otherwise, it is unstable.
- Frequency Response: The positions of poles and zeros will give insights into the system's behavior in the frequency domain, with poles close to the unit circle potentially leading to high-frequency resonance and zeros affecting the frequency attenuation.

This manual plotting method helps you understand the construction of pole-zero plots without relying on built-in functions, providing a deeper understanding of the underlying mathematics in signal processing and control systems.

Experiment 10.2

Z-Transform and Inverse Z-Transform

Aim:

Compute symbolic Z-transform and its inverse.

Program:

```
% Test Case: Numerical Z-Transform and Inverse Z-Transform
clc;
clear;
close all;
% ======================
% Numerical Z-Transform
% ======================
% Given sequence
x = [1 0.5 0.25 0.125];
```

```
% Number of points
N = 64;
% Time index
n = 0:length(x)-1;
% Define z-points on the unit circle
k = 0:N-1;
omega = 2*pi*k/N;
% Unit circle points
z = exp(1i*omega);
% Compute Z-transform on unit circle
Xz = zeros(1,N);
for m = 1:N
Xz(m) = sum(x .* (z(m).^(-n)));
end
% =====================
% Plot Magnitude and Phase
% =====================
figure;
subplot(2,1,1);
plot(omega/(2*pi), abs(Xz), 'b', 'LineWidth', 1.5);
xlabel('Normalized Frequency');
ylabel('|X(z)|');
title('Magnitude Spectrum');
grid on;
```

```
subplot(2,1,2);
plot(omega/(2*pi), angle(Xz), 'r', 'LineWidth', 1.5);
xlabel('Normalized Frequency');
ylabel('Phase (rad)');
title('Phase Spectrum');
grid on;
% ======================
% Numerical Inverse Z-Transform
% ======================
% Use inverse DFT to recover x[n]
x_rec = zeros(1,N);
for m = 1:length(n)
x_rec(m) = (1/N) * sum(Xz .* (z.^n(m)));
end
% Display recovered sequence
disp('Original sequence:');
disp(x);
disp('Recovered sequence from Inverse Z-Transform:');
disp(real(x_rec(1:length(x))));
```

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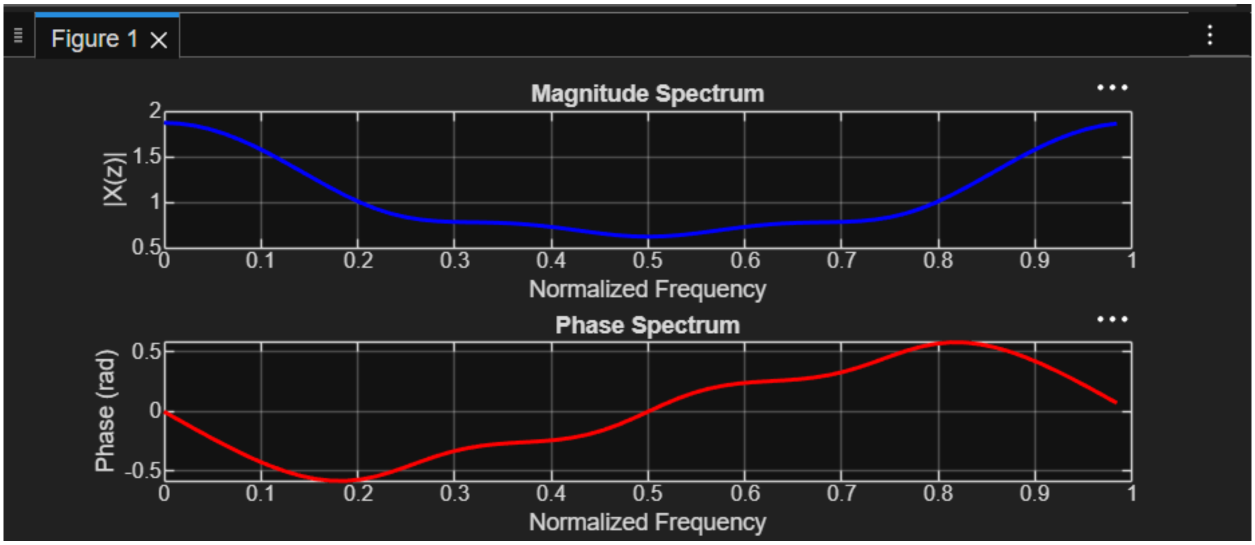
Output Generated Graph:



Result:

Z-transform derived for given sequence.